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INDUSTRY PROJECT PHASE 2

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PART A: Presentation of the Information System Solution

(Improving the Business Processes at Vhutambo Edu Play)

1. Introduction to the Proposed Solution

The Information System solution was designed to address inefficiencies identified during the business process analysis phase.
The solution focuses on **automation, accuracy, and efficiency** within Vhutambo Edu Play.

The proposed system is an **Integrated Retail Information System** consisting of:

- A **Point of Sale (POS) System**
- An **Inventory Management System**
- A **Basic Business Website**

2. Business Processes Improved by the System

2.1 Sales Process Improvement

Before (AS-IS):

- Manual price calculation
- Handwritten sales records
- No daily sales summary

After (TO-BE):

- Automated price calculation
- Digital sales records
- Daily and monthly sales reports

Impact: Faster transactions, fewer errors, improved customer service.

2.2 Inventory Management Improvement

Before (AS-IS):

- Manual stock counting
- No real-time stock visibility
- Frequent stock shortages

After (TO-BE):

- Automatic stock updates after each sale
- Low-stock alerts
- Accurate inventory records

Impact: Reduced losses, better stock planning, improved productivity.

2.3 Reporting and Decision-Making

Before (AS-IS):

- Manual calculations
- No performance trends
- Limited business insight

After (TO-BE):

- Automated sales and inventory reports
- Profit tracking
- Data-driven decision-making

Impact: Improved management control and profitability monitoring.

3. Website as a Business Support Tool

The business website supports:

- Online visibility
- Product awareness
- Customer communication

Although not a full e-commerce platform, it provides a **foundation for future growth**.

4. Overall Benefits of the Information System

- Increased operational efficiency
- Improved accuracy
- Reduced manual workload
- Enhanced customer experience
- Improved profitability control

PART B: Final Project Management Principles and Methodologies Applied

(During Implementation of the Information System)

5. Project Management Methodology Used

The **Waterfall Methodology** was applied due to:

- Clearly defined requirements
- Structured phases
- Suitability for small-scale IT projects

6. Project Lifecycle Phases Applied

6.1 Initiation Phase

- Identification of business problems
- Definition of project objectives
- Stakeholder identification (Shop owner, staff, intern)

***Principle Applied*:** Clear project justification

6.2 Planning Phase

- Scope definition
- Resource planning
- Project scheduling
- Risk identification

Principle Applied: Proper planning ensures project success

6.3 Design Phase

- System architecture design
- Process redesign (AS-IS → TO-BE)
- User interface planning

***Principle Applied*:** User-focused system design

6.4 Development Phase

- POS system setup
- Inventory system configuration
- Website development

Principle Applied: Controlled execution according to plan

6.5 Testing Phase

- System functionality testing
- Accuracy of inventory updates
- Sales transaction testing

Principle Applied: Quality assurance and control

6.6 Deployment Phase

- System installation
- User orientation and training
- System handover

Principle Applied: Smooth transition and stakeholder readiness

6.7 Closure Phase

- Final documentation
- Evaluation of project outcomes
- Lessons learned

Principle Applied: Formal project closure and accountability

7. Project Management Principles Applied

Principle	Application
Scope Management	Clearly defined system boundaries
Time Management	Phased timeline planning
Risk Management	Identified and mitigated risks
Quality Management	Testing before deployment
Stakeholder Management	Continuous engagement

8. Risk Management During Implementation

Risk	Impact	Mitigation
Resistance to system use	Medium	User training
Limited budget	High	Use of open-source tools
Technical challenges	Medium	Simple system design

9. Monitoring and Control

- Progress tracking against milestones
- Review meetings
- Issue resolution

***Principle Applied*:** Continuous monitoring ensures alignment with objectives

10. Conclusion

Phase 2 demonstrates the successful application of **project management principles and methodologies** in delivering an Information System solution that improves business processes at Vhutambo Edu Play.
The structured approach ensured the system addressed real business needs while remaining practical and sustainable.